



COPILOTTRAINING
TECHNICAL BRIEFING

No Instruments, No Delivery

| The Enterprise Agentic Imperative

Agentic AI multiplies developer throughput — but only if you build the cockpit first.

Executive Talk · 45 minutes



A Three-Part Executive Briefing Series



The Labor Multiplier

What agents can actually do — the 67% of delivery labor outside the code editor that AI is built to handle at a fraction of human cost



Agentic Economics

How to calculate and capture ROI — agent compute costs, infrastructure investment requirements, and the honest payback math



Agentic Delivery

The operating model for safe scale — instrument infrastructure, pilot oversight, and the governance that enables the throughput multiplier



Each briefing is complete on its own. Together they form a full strategic framework for agentic AI investment.

Three Briefings, Three Decisions

The Labor Multiplier

Which of the 67% of delivery labor is ready to automate — and what organizational investment does capturing it require?

Opportunity Decision

Agentic Economics

What is the realistic ROI timeline, and what infrastructure makes the \$2–5/hour agent arbitrage accessible at scale?

Capital Decision

Agentic Delivery

What governance infrastructure — instruments, oversight model, no-fly zones — must exist before agent programs can operate safely?

Operating Model Decision

These are the three decisions the briefing series is designed to drive. This talk addresses one of them.

The Complete Decision Framework



Identify the Opportunity

The Labor Multiplier: 67% of delivery labor has no AI coverage — discovery, compliance, documentation, and validation are ready for agents



Quantify the Investment

Agentic Economics: agent compute at \$2–5/hour creates a structural cost arbitrage — the ROI depends on the verification infrastructure



Enable Safe Operation

Agentic Delivery: the instrument panel, pilot model, and governance structure that make the multiplier achievable at enterprise scale



The Labor Multiplier identifies what is possible. Agentic Economics quantifies the return. Agentic Delivery specifies the infrastructure that makes both achievable.

PART 1

The Shift

From Coders to Captains

Business stakes and the reframe that changes the calculus for enterprise AI investment



PART 2

The Flight Model

Roles, Instruments, Multiplier

The operating model in three slides — and the math that makes it worth building

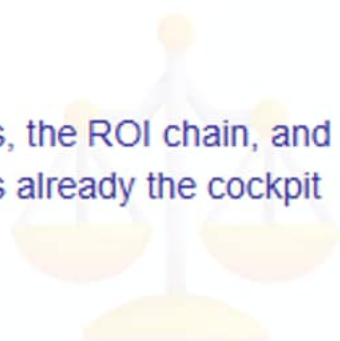


PART 3

The Risk

Liability Charter

Six governance categories, the ROI chain, and why mature DevSecOps is already the cockpit



PART 4

The Decision

Three Named Asks

Named owners, timelines, and exactly what leadership is authorizing today



The Shift

The productivity opportunity is documented — the question is what it takes to capture it safely at enterprise scale



The Signal

Only 5% of AI pilots deliver material bottom-line improvement



The Stakes

\$4.88M breach + SEC 4-day disclosure — liability startups avoid



The Reframe

Developers become captains — throughput follows the instruments

`executive briefing`

`↳ AI is already changing software delivery at measurable scale. The question is whether an organization captures it safely.`

The 5% Signal

Only 5% of enterprise AI pilots deliver material bottom-line improvement. Governance infrastructure is the differentiator.

5%

of enterprise AI pilots deliver material bottom-line improvement

McKinsey QuantumBlack, State of AI 2024

✓ The RCT Evidence

GitHub 2022: 55% faster with AI. Accenture 2024: 84% more successful builds at enterprise scale.

🔧 What Separates the 5%

Automated verification infrastructure — the instrument panel that makes AI trustworthy at speed.

💡 The Implication

Governance infrastructure is the variable. Organizations that invest in it capture the full gain.

🔗 Organizations capturing AI value built the infrastructure to verify what the tools produce.

The Enterprise Stakes



DORA 208× Gap

Elite teams deploy 208× more frequently with 3× lower failure rate — DORA 2023.



\$4.88M Breach Cost

IBM 2024: average enterprise data breach cost. AI-assisted detection saves \$2.2M per incident.



Regulatory Floor

SEC 4-day disclosure in force. EU AI Act 2025–2026. NIST AI RMF is the governance standard.



Agentic AI without automated governance creates a new delivery liability category. The instrument panel is both the multiplier enabler and the liability defense.

From Coders to Captains



Developer as Coder

Lines of code written

Typing speed

Syntax knowledge

Faster autocomplete



Developer as Captain

Missions completed

Instrument monitoring capacity

Judgment and decision-making

Autopilot — developer remains accountable

⚡ One developer supervising 3–4 concurrent agentic sessions converts a \$10M organization into \$20M of delivery capacity. The Flight Model shows the math.

The Flight Model

Five roles, six instruments, and the authorization model that makes the multiplier achievable



Five Roles

FAA to Autopilot — every enterprise role has a mapped position



Six Instruments

Automated governance gates — each maps to an enterprise liability



The Math

One developer, 3–4 sessions — \$20M capacity in a \$10M org

`flight model`

`↳ Instrument monitoring capacity is the ceiling. The instrument panel determines how many flights are safe.`

The Enterprise Flight System

Five roles — every organization already has most of them

FAA

Sets all rules — safety standards, enforcement, compliance

CISO · Legal · Compliance

Control Tower

Coordinates active flights — sequencing, routing, go/no-go decisions

CTO · VP Engineering

Ground Crew

Prepares instruments and infrastructure for every flight

Platform Engineering · DevOps

Captains

Plans the mission, makes judgment calls, lands safely, deploys code

Developers

Autopilot

Executes the mechanical work — code generation, test iteration, docs

Agentic AI

Most organizations have Captains and Autopilot. Ground Crew and the instrument panel are the missing components.

The Instrument Panel

Six governance gates — each provides a single traffic-light reading at deployment

Test Health	Automated test results gate every deployment — red grounds the flight	Quality gate
Security Scan	Vulnerability detection on every commit — \$4.88M breach defense	SAST / DAST
Performance	Regression detection vs. production baseline — silent slowdowns caught	Baseline guard
Compliance	SOC 2, HIPAA, FedRAMP violations blocked at merge — not in audit	Policy gate
Deploy Window	Change management enforced automatically — no override, no exceptions	Risk guard
Dependencies	Supply chain CVE scan on every import — SolarWinds-class risk detected	CVE gate

Instrument coverage is the prerequisite for safe AI output verification at speed. The multiplier follows from the instrument panel.

Authorization and the Multiplier

The Authorization Model

Human defines scope, acceptance criteria, limits — go/no-go decision

Human initializes agent with bounded parameters and explicit constraints

AI executes autonomously — developer monitors instrument panel

Human reviews all output — all six instruments green before deploying

The Multiplier

One developer monitors aggregated status across all active flights

Throughput equivalent to tripling headcount without tripling payroll

Not typing speed — the instrument panel determines how many flights are safe

 AI operates in one bounded phase. Humans control entry and exit. That authorization model is what makes 3–4 concurrent sessions governable at enterprise scale.

The Risk

Six liability categories, the ROI chain, and why existing DevSecOps infrastructure is already the cockpit



Governance Charter

Six AI actions requiring authorization
— each with a dollar figure



The CFO and CISO Case

\$2.2M savings + \$1→\$6 ROI + EU AI
Act and SEC timelines



The Instrument Advantage

Mature DevSecOps orgs are
completing a cockpit, not building one

`governance`

`↳ The instrument panel is flight clearance.`

The Governance Charter

Six categories of AI action requiring explicit human authorization — each with a liability citation

DB Schema	Schema changes are irreversible at scale — creates breach exposure	\$4.88M risk
Security Ctrl	Bypassing a compliance check can trigger SEC 4-day disclosure	SEC regulatory
Dependencies	Supply chain trust — SolarWinds and Log4j establish the precedent	Supply chain
Prod Config	Runtime config shifts behavior as dramatically as code changes	Deploy risk
Access Ctrl	Privilege escalation breaks the trust model for automation	Audit flag
Integrations	New external connections need legal, security, and compliance sign-off	Architecture

These limits earn the agent broad autonomous authority within them. Governance boundaries are what make autonomy safe.

The CFO and CISO Case

The CFO Frame

IBM 2024 — before SEC disclosure obligations and regulatory fines

IBM 2025 — with AI-assisted security detection in the instrument panel

Every dollar in delivery automation returns \$6 in reduced incident costs

Protecting the ROI on AI spend already authorized

The CISO Frame

High-risk AI classification and conformity requirements — build now or build under regulatory audit

Material cybersecurity incidents require disclosure within 4 business days

Govern, Map, Measure, Manage — the enterprise AI governance vocabulary regulators use

Organizations with existing automation are completing an instrument panel — a structural advantage.

 Building governance infrastructure on the organization's own terms and timeline is the lower-cost path. Regulatory pressure makes it more expensive.

The Decision

Three named asks, three owners, three timelines — what is being authorized today



Authorize the Audit

VP Engineering. Four weeks. Every gap is risk with open eyes.



Fund the Cockpit

3–5 platform engineers — enabling infrastructure, not overhead.



Designate Pilots

One team. 90 days. Baseline DORA metrics defined before launch.

authorization

↳ Authorization here is an operating model decision, with named owners and a 90-day proof structure.

Action 1: Authorize a Platform Audit



What

Assess which of the six instruments the pipeline has. Each gap is a named ceiling on AI ROI.



Owner: VP Engineering

Leads the assessment with the delivery team. Results reported to leadership in four weeks.



Timeline: 4 Weeks

Every gap is risk accepted with open eyes — a ceiling on the AI investment already authorized.



This is the only ask with a hard deadline. Actions 2 and 3 depend on what this one finds.

Action 2: Fund the Instrument Panel



What

A dedicated platform team wiring existing tooling and filling gaps identified in the audit.



Investment: 3–5 Engineers

For a 50-developer organization. Scales to 500. Payback measured in weeks, not quarters.



Return: \$1 → \$6

\$1 in delivery automation returns \$6 in reduced incident costs — IBM/Ponemon.



The platform team is the capital decision that makes existing AI spend return value.

Action 3: Designate a Pilot Team



What

Select one team to operate with a complete instrument panel and structured agentic workflow.



Scope: 90-Day Pilot

Measure deployment frequency, failure rate, and cycle time — baseline before launch, data after.



Output: The Business Case

90-day DORA-anchored data is the board-ready case for scaling agentic delivery organization-wide.



Authorization here is an operating model decision. Start with one team. Measure everything. Scale what works.

The Instrument Panel Changes Everything

Before

One developer, one agentic session — manual verification at every step

AI-generated code ships without automated security or compliance checks

Flying blind across more missions amplifies failures, not productivity

Risk is invisible until incident — SEC 4-day clock starts after the breach

After

One developer, 3–4 concurrent sessions with aggregated instrument status

Automated gates on every commit — security, compliance, quality enforced

Fleet-wide visibility — CTO/VP Engineering coordinates from the control tower

Speed compounds advantage safely — 208× deployments, 3× lower failure rate

3–4×

developer throughput with a full instrument panel

\$20M

delivery capacity unlocked in a \$10M engineering org

6×

ROI on every dollar invested in delivery automation

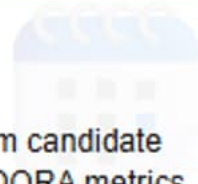
✓ WHAT YOU CAN DO TODAY

Today



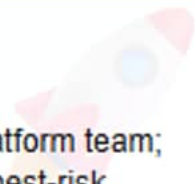
- Schedule the platform infrastructure audit with VP Engineering — four-week deadline, instrument gap map as the deliverable
- Identify which of the six instruments the delivery pipeline already has

This Week



- Designate a pilot team candidate and define baseline DORA metrics before agentic workflow begins
- Draft the platform team investment proposal — 3–5 engineers, specific gaps identified, payback timeline

This Month



- Fund and staff the platform team; begin closing the highest-risk instrument gaps first
- Pilot team completes first full agentic sprint with complete instrument monitoring active

🔑 Key Takeaway

The organizations that lead are those whose developers can safely fly the most planes.

 AI Productivity Research[GitHub / Microsoft Research, 2022](#)

Controlled study: developers complete tasks 55% faster with AI coding assistance

[GitHub + Accenture RCT, 2024](#)

Enterprise scale: 84% more successful builds; 70% less mental effort on repetitive tasks

[McKinsey QuantumBlack: State of AI, 2024](#)

Only 5% of AI pilots deliver material bottom-line improvement; organizational complexity is the primary barrier

[METR RCT, 2025](#)

Experienced developers took 19% longer on complex tasks with AI assistance — expected 20–39% speedup

 Risk, Governance and Regulation[IBM Cost of a Data Breach, 2025](#)

\$4.88M average breach cost; AI-assisted detection saves \$2.2M per incident

 DevOps and Measurement[DORA State of DevOps, 2023](#)

Elite teams deploy 208× more frequently with a 3× lower change failure rate

[DORA: ROI of AI-Assisted Development, 2025](#)

Framework linking AI engineering velocity to measurable financial outcomes



No Instruments, No Delivery

THE ENTERPRISE AGENTIC IMPERATIVE



5%

Deliver Material ROI

McKinsey 2024: the differentiator is governance infrastructure, not the tools



\$20M

Capacity in a \$10M Org

One developer, 3–4 concurrent sessions
— limit is monitoring, not headcount



\$2.2M

Per-Incident Savings

IBM 2025: AI-assisted security detection reduces breach cost by \$2.2M per event

Which of the six instruments is the most critical gap in the delivery pipeline right now?